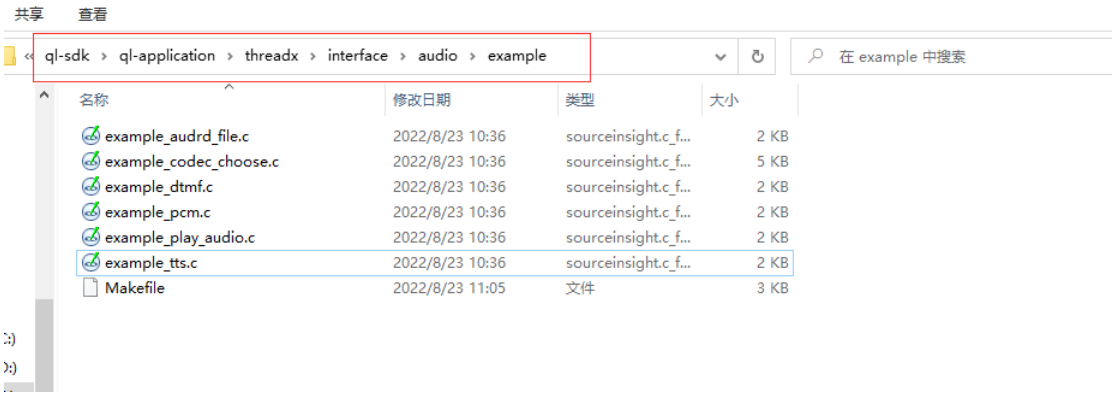
Author: Felix Liu(刘炎铭) Felix.liu@quectel.com
日期：2022 年 8 月 31 日

下面是基于 LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK 版本进行的 TTS 功能的添加操作

1. 添加 TTS 所在外设文件到 Makefile 文件中

   Example_tts.c 文件所在路径



   并修改 example_tts.c 文件

```
static void quec_tts_test(void * argv)
{
    //等待audio初始化好
    while(!ql_get_audio_state()){
        ql_rtos_task_sleep_ms(50);
    }

    printf("quec_tts_test start ...\r\n");

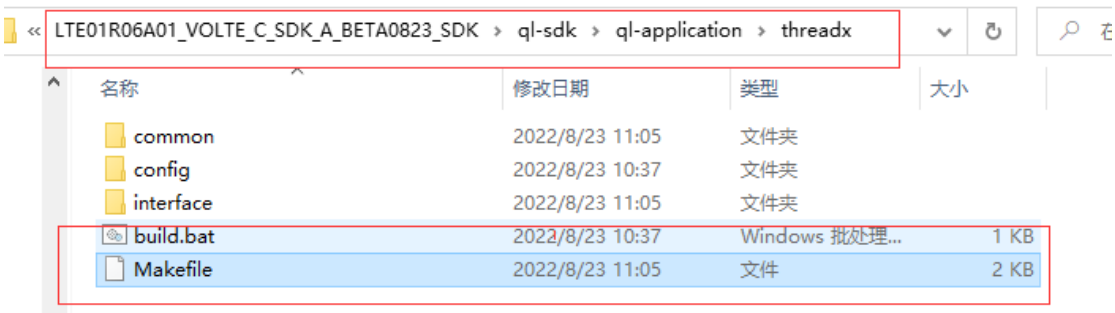
    ql_set_audio_path_receiver();

    ql_codec_choose(AUD_INTERNAL_CODEC, NULL);
    ql_tts_init(test_cb);
    ql_set_volume(8);

    ql_tts_set_speed(0);                                    //设置tts播放速度，参数范围-32768~32767
    ql_tts_set_volume(0);                                   //设置tts音量，参数范围-32768~32767
    ql_tts_digit_mode(0);                                   //数字处理策略，0自动判断，1数字作号码处理，2数字作
    while(1)
    {
        ql_tts_play(2,"hello,123欢迎使用移远模块");              //mode2,使用GBK编码方式
        ql_rtos_task_sleep_s(5);                            //等待5S,由于tts为异步播放，连续调用会直接打断
        ql_tts_play(1,"6B228FCE4F7F752879FB8FDC6A215757");  //mode1,使用UNICODE-16编码方式
        ql_rtos_task_sleep_s(5);                            //等待5S
    }
}
//application_init(quec_tts_test, "quec_tts_test", 2, 0);
```

解除掉注释

   Makefile 所在路径



   将 TTS 相关文件的路径按照格式添加进 Makefile 文件。

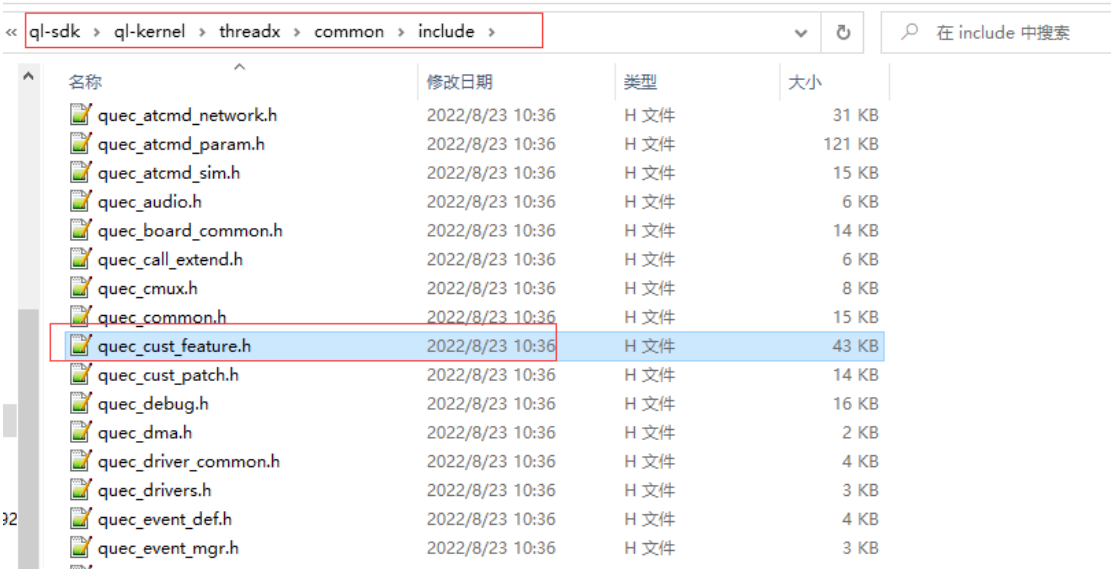
```
#-------------------------------------------------------------------------------
# Configure variable
#-------------------------------------------------------------------------------
ifneq (${TOP_DIR},)
  TOP_DIR:=${TOP_DIR}
else
  TOP_DIR:=${CURDIR}
endif


#-------------------------------------------------------------------------------
# Configure compile directories
#-------------------------------------------------------------------------------
COMMPILE_DIRS:= \
  interface/at/src \
  interface/log/src \
  interface/audio/example\


#-------------------------------------------------------------------------------
# Configure default libarays
#-------------------------------------------------------------------------------
DEFAULT_LIBS+=  \
  common/lib/ql_common_api.lib \
  common/lib/com_cell_locator.lib \
  common/lib/com_ftp.lib \
  common/lib/com_http.lib \
```

修改完 Makefile 文件下保存。

在下面目录下找到 quec_cust_feature.h 文件进行修改



```
#define QUEC_WDTMF_SUPPORT        (1)
//Egan.li Add DTMF detection support
#define QUEC_DTMF_DETECT_SUPPORT  (1)
#define QUEC_DTMF_LOCTED_SUPPORT  (1)
//Egan.li Add LTONE support
#define QUEC_LTONE_SUPPORT        (1)

//Egan.li 20200616: add tone incoming support
#define QUEC_AUDIO_TONE_INCOMING_SUPPORT (1)


#ifndef  QUEC_TTS_NOT_SUPPORT

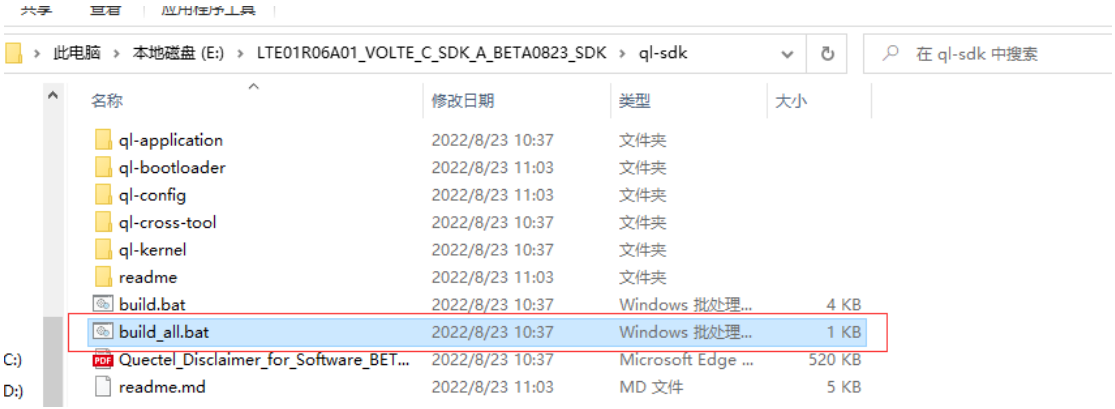
#ifdef QUECTEL_OPEN_CPU
#define QUEC_AUDIO_TTS_SUPPORT (1)
#else
#define QUECTEL_TTS_FEATURE
//Fitz.xu 20200401: add tts task XIP support
#define QUEC_TTS_TASK_XIP_SUPPORT
//Egan.li add WTTS support
#define QUEC_QWTTS_SUPPORT (1)
//Egan.li add 16K TTS support
#define QUEC_TTS_SUPPORT_IFK  (1)
#endif
#endif

#ifndef QUEC_AMR_NOT_SUPPORT
//Egan.li amr support
#define QUEC_AUDIO_AMR_SUPPORT    (1)
```

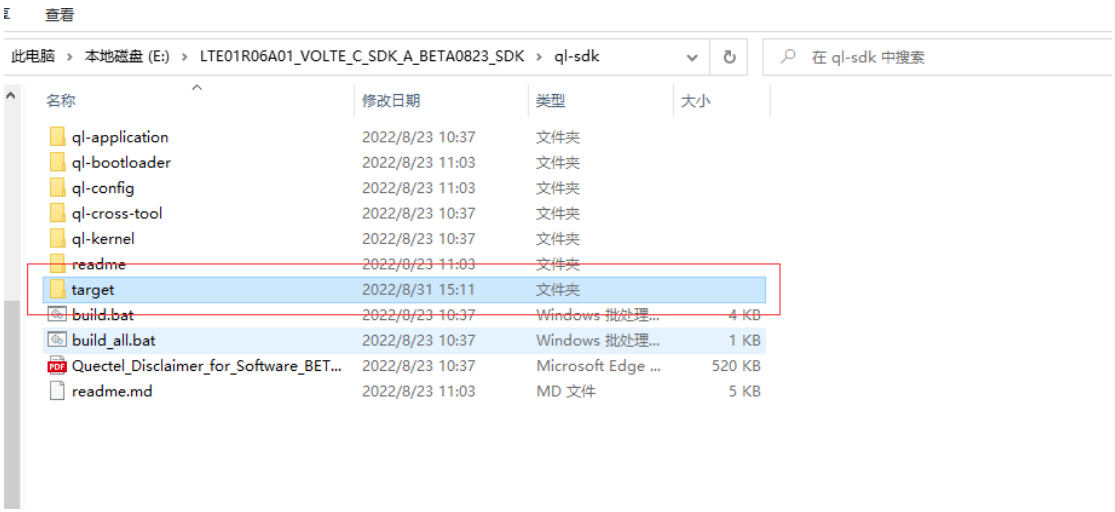
将此项改为#if 1

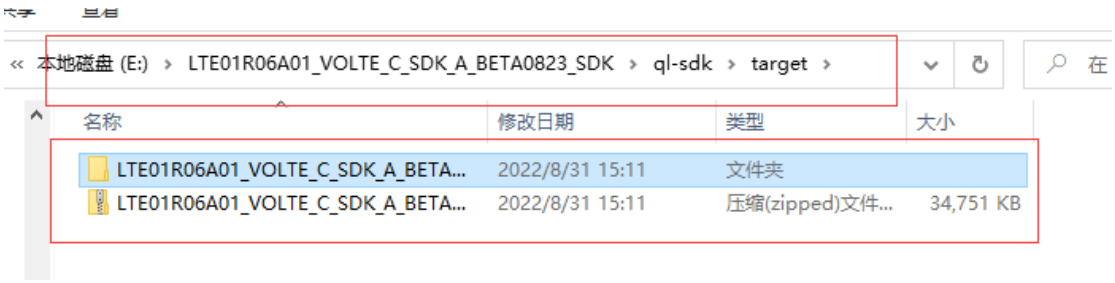
VOLTE 版本是多合一版本，在 build all.bat 之前需要先根据自己设备去编译以下指令（例如我选择的设备是 EC800MCN_GA，则.\build.bat project_name EC800MCN_GA +自定义的版本号）

2. 双击 build_all.bat

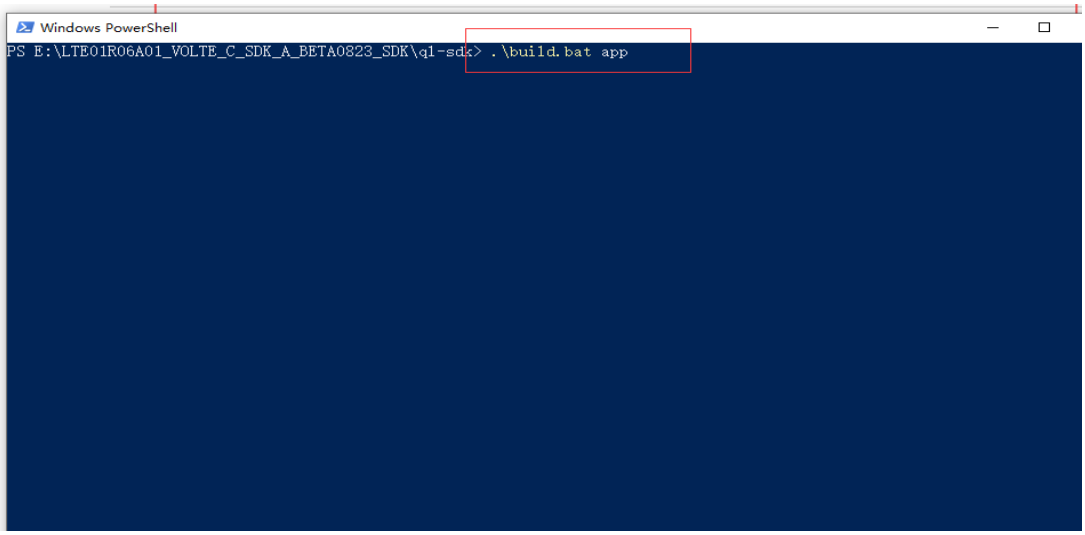


编译完成后会出现个 target 文件





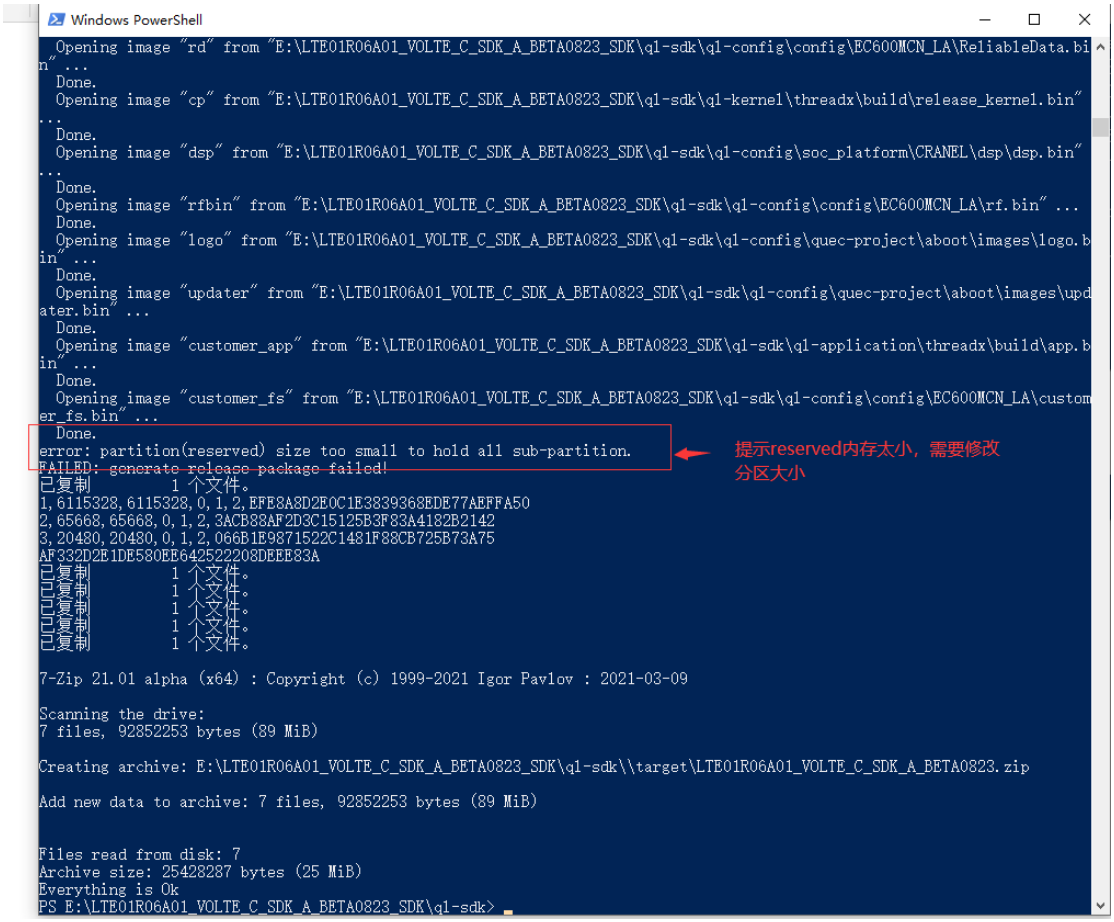
3.进行编译.\build.bat app



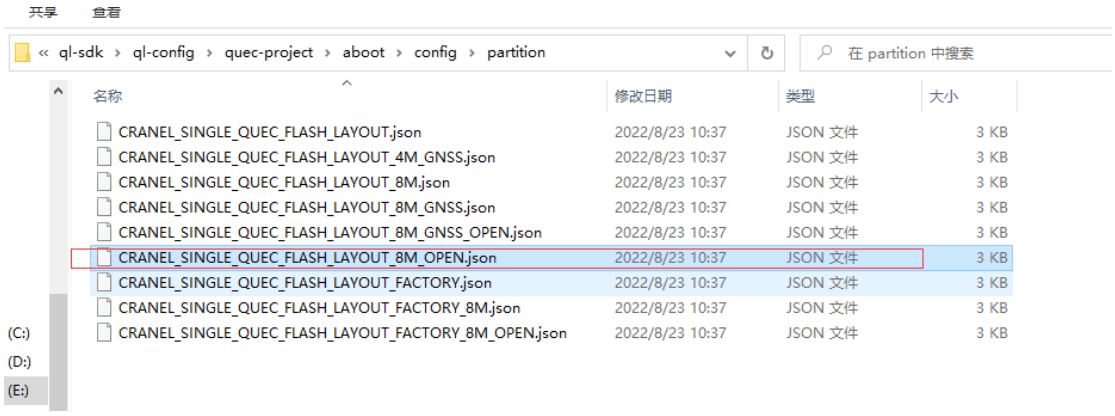
```
删除文件 - E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\build\app.bin
删除文件 - E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\build\app.elf
删除文件 - E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\build\app.map
-----------------compile [interface/at/src]--------------------
gnumake -C interface/at/src -f Makefile TOP_DIR=E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\
gnumake[1]: Entering directory `E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK/ql-sdk/ql-application/threadx/interface/at/src
'
gnumake[1]: Leaving directory `E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK/ql-sdk/ql-application/threadx/interface/at/src'
-----------------compile [interface/log/src]--------------------
gnumake -C interface/log/src -f Makefile TOP_DIR=E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx
\
gnumake[1]: Entering directory `E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK/ql-sdk/ql-application/threadx/interface/log/sr
c'
gnumake[1]: Leaving directory `E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK/ql-sdk/ql-application/threadx/interface/log/src
'
-----------------compile [interface/audio/example]--------------------
gnumake -C interface/audio/example -f Makefile TOP_DIR=E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\t
hreadx\
gnumake[1]: Entering directory `E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK/ql-sdk/ql-application/threadx/interface/audio/
example'
gnumake[1]: Leaving directory `E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK/ql-sdk/ql-application/threadx/interface/audio/e
xample'
--------------link [build/app.elf]----------------------
E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\/../..\ql-cross-tool\win32\owtoolchain\gcc-arm-n
one-eabi\bin\arm-none-eabi-ld.exe -gc-sections   -TE:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threa
dx\\config\common\app_linkscript.ld -nostdlib  -Map build/app.map -o build/app.elf --whole-archive   E:/LTE01R06A01_VOLT
E_C_SDK_A_BETA0823_SDK/ql-sdk/ql-application/threadx/interface/at/src//build/src.lib  E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0
823_SDK/ql-sdk/ql-application/threadx/interface/log/src//build/src.lib  E:/LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK/ql-sdk
/ql-application/threadx/interface/audio/example//build/example.lib   common/lib/ql_common_api.lib common/lib/com_cell_lo
cator.lib common/lib/com_ftp.lib common/lib/com_http.lib common/lib/com_mbedtls.lib common/lib/com_paho_mqtt.lib common/
lib/com_qthsdk.lib common/lib/com_ssl.lib  --no-whole-archive  E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-appli
cation\threadx\/../..\ql-cross-tool\win32\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7\nofp\libc.a E:\LTE01R
06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\/../..\ql-cross-tool\win32\owtoolchain\gcc-arm-none-eabi\
arm-none-eabi\lib\thumb\v7\nofp\libstdc++.a E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\/../
..\ql-cross-tool\win32\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7\nofp\libg.a E:\LTE01R06A01_VOLTE_C_SDK_A
_BETA0823_SDK\ql-sdk\ql-application\threadx\/../..\ql-cross-tool\win32\owtoolchain\gcc-arm-none-eabi\lib\gcc\arm-none-ea
bi\9.2.1\thumb\v7\nofp\libgcc.a E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\/../..\ql-cross-
tool\win32\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7\nofp\libnosys.a E:\LTE01R06A01_VOLTE_C_SDK_A_BETA082
3_SDK\ql-sdk\ql-application\threadx\/../..\ql-cross-tool\win32\owtoolchain\gcc-arm-none-eabi\arm-none-eabi\lib\thumb\v7\
nofp\libm.a
--------------objcopy [build/app.bin]-------------
E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\/../..\ql-cross-tool\win32\owtoolchain\gcc-arm-n
one-eabi\bin\arm-none-eabi-objcopy.exe  -O binary build/app.elf  build/app.bin
E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-application\threadx\/../..\ql-cross-tool\win32\host\tools\crc_bin.ex
e build/app.bin
-------------------------------------------------------
  GCC Compiling Finished Sucessfully.
  The target image is in the 'build' directory.
-------------------------------------------------------
PS E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk>
```

继续编译.\build.bat firmware

```
  The target image is in the 'build' directory.
-------------------------------------------------------
PS E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk> .\build.bat firmware
```



4.修改分区大小（通过减小 customer_app 分区大小来增大 reserved 的大小若是依然显示）
LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\ql-config\quec-project\aboot\config\partition\CRANEL_SINGLE_QUEC_FLASH_LAYOUT_8M_OPEN.json
（基于 EC800M_GA 则修改 CRANEL_SINGLE_QUEC_FLASH_LAYOUT_8M__GNSS_OPEN.json）
在此目录下找到此文件夹进行修改



```
    },
    {
      "name": "reserved",
      "format": "raw",
      "size": "-1"
    },
    {
      "name": "customer_app",
      "format": "raw",
      "size": "500KiB"
    },
    {
      "name": "customer_fs",
      "format": "raw",
      "size": "1024KiB"
    },
      {
      "name": "fota_pkg",
```

可将500改为100，以调整分区大小

重新编译

```
Everything is Ok
PS E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk> .\build.bat firmware
```

```
Windows PowerShell
      Done.
      Using already opened/generated image "fwcerts" with id "fwcerts.bin" ...
      Done.
      Using already opened/generated image "rd" with id "ReliableData.bin" ...
      Done.
      Using already opened/generated image "cp" with id "cp.bin" ...
      Done.
      Using already opened/generated image "dsp" with id "dsp.bin" ...
      Done.
      Using already opened/generated image "rfbin" with id "rf.bin" ...
      Done.
      Using already opened/generated image "logo" with id "logo.bin" ...
      Done.
    Done.
  Done.
Done.
Calculating total progress weight ...
Done.
Generating download commands ...
Done.
Generating crane firmware image ...
Done.
Generating target release package ...
Done.
Release package generated successfully!
已复制        1 个文件。
1,6115328,6115328,0,1,2,EFE8A8D2E0C1E3839368EDE77AEFFA50
2,65668,65668,0,1,2,3ACB88AF2D3C15125B3F83A4182B2142
3,20480,20480,0,1,2,066B1E9871522C1481F88CB725B73A75
AF332D2E1DE580EE642522208DEEE83A
已复制        1 个文件。
已复制        1 个文件。
已复制        1 个文件。
已复制        1 个文件。
已复制        1 个文件。

7-Zip 21.01 alpha (x64) : Copyright (c) 1999-2021 Igor Pavlov : 2021-03-09

Scanning the drive:
8 files, 105552951 bytes (101 MiB)

Creating archive: E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk\\target\LTE01R06A01_VOLTE_C_SDK_A_BETA0823.zip

Add new data to archive: 8 files, 105552951 bytes (101 MiB)


Files read from disk: 8
Archive size: 38124602 bytes (37 MiB)
Everything is Ok
PS E:\LTE01R06A01_VOLTE_C_SDK_A_BETA0823_SDK\ql-sdk>
```

还要注意一点，关注 customer_app 的地址，通过加载固件到 aboot 工具后查看 customer_app 的地址，此时这个 flash 的地址和大小变动之后，ql-application\threadx\config\common\app_linkscript.ld 文件中的_ROM_START_ADDR 的地址要与 aboot 里 customer_app 的地址保持一致，否则 app 运行将会出错。

```
__ROM_START_ADDR = 0x8063B000;
__MAX_ROM_SIZE = 0xDA8000;

__RAM_START_ADDR = 0x7e420000;
__MAX_RAM_SIZE   = 0x100000;

MEMORY
{
    ROM (rx)      : ORIGIN = __ROM_START_ADDR + 8, LENGTH = __MAX_ROM_SIZE
    SRAM (rwx)      : ORIGIN = __RAM_START_ADDR, LENGTH = __MAX_RAM_SIZE
}

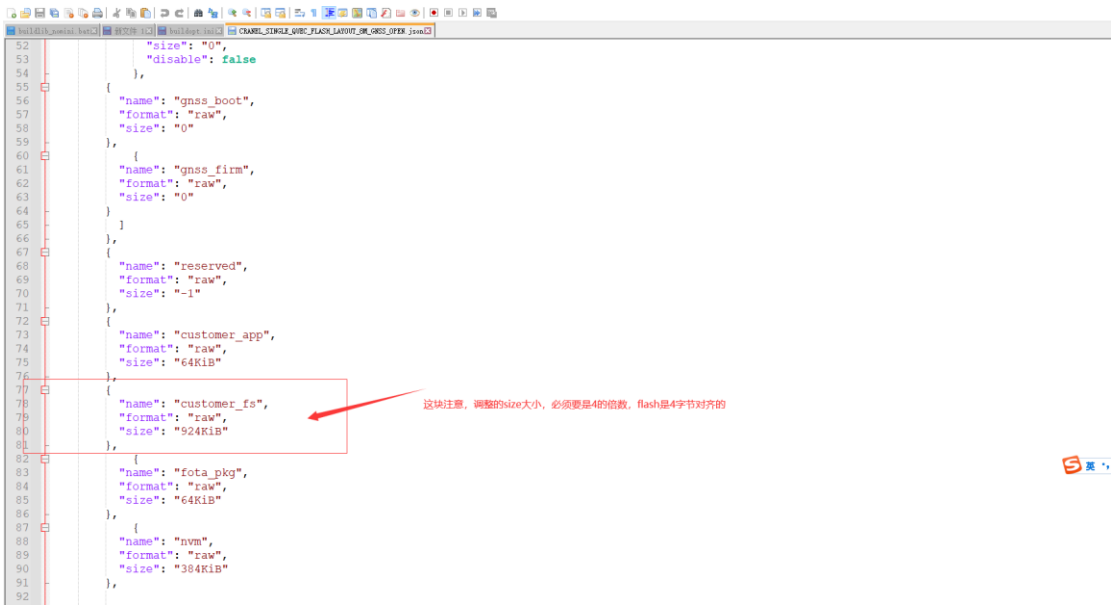
SECTIONS
{
    . = __ROM_START_ADDR + 8;
    .initdata : ALIGN(4)
    {
        KEEP(*(.initdata))
    }>ROM
    .userStartUp :ALIGN(4)
    {
        KEEP(*(.userInit))
    }>ROM
    .text : ALIGN(4)
    {
        *(.text)
        *(.text*)
    }>ROM
    .rodata : ALIGN(4)
    {
        *(.rodata)
        *(.rodata*)
```

Partition table

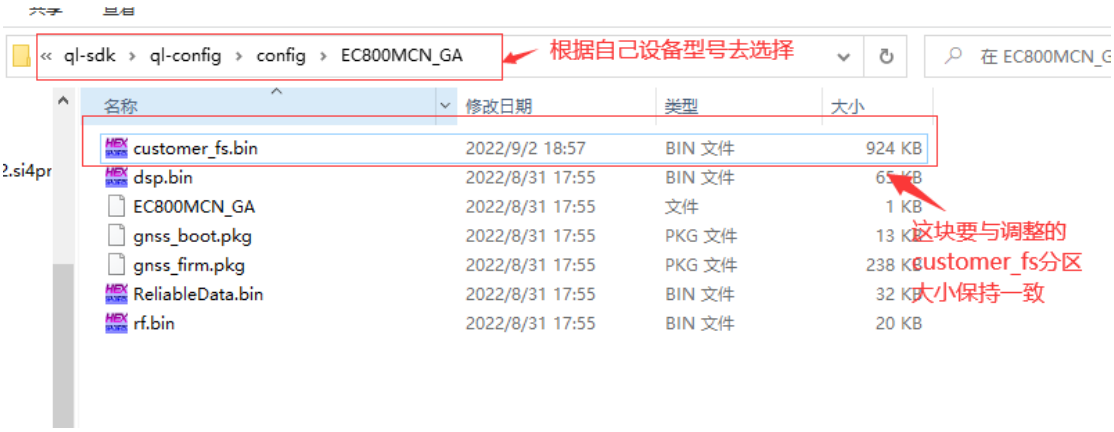
Name	Type	Start	Size	vStart	vSize
all	group	0x00000000	0x00800000	0xFFFFFFFF	0x00800000
qspi	flash	0x00000000	0x00800000	0x80000000	0x00800000
bootloader	ubi	0x00000000	0x00016000	0x80000000	0x00016000
system	part	0x00016000	0x005FD000	0x80016000	0x005FD000
ptable	raw	0x00016000	0x00001000	0x80016000	0x0000088C
fwcerts	raw	0x00017000	0x00003000	0x80017000	0x000007F4
rd	raw	0x0001A000	0x00008000	0x8001A000	0x00008000
cp	raw	0x00022000	0x005D5000	0x80022000	0x005D5000
dsp	raw	0x005F7000	0x00011000	0x805F7000	0x00010084
rfbin	raw	0x00608000	0x00005000	0x80608000	0x00005000
logo	raw	0x0060D000	0x00006000	0x8060D000	0x00005BB6
reserved	raw	0x00613000	0x00028000	0x80613000	0x00028000
customer_app	raw	0x0063B000	0x00019000	0x8063B000	0x000068B0
customer_fs	raw	0x00654000	0x00100000	0x80654000	0x00100000
fota_pkg	raw	0x00754000	0x00010000	0x80754000	0x00010000
nvm	raw	0x00764000	0x00060000	0x80764000	0x00060000
erase_rd	raw	0x007C4000	0x00001000	0x807C4000	0x00001000
updater	raw	0x007C5000	0x00020000	0x807C5000	0x000178F8
fota_param	raw	0x007E5000	0x00003000	0x807E5000	0x00003000
quec_cfg	raw	0x007E8000	0x00008000	0x807E8000	0x00008000
factory	part	0x007F0000	0x00010000	0x807F0000	0x00010000
factory_a	raw	0x007F0000	0x00008000	0x807F0000	0x00008000
factory_b	raw	0x007F8000	0x00008000	0x807F8000	0x00008000

然后依次编译 .\build.bat app、.\build.bat firmware 即可。

若是调整 customer_app flash 分区的大小依然还是存在 reserved to small 的问题，可以适当去调整 customer_fs flash 分区的大小，同样在以下文件去修改 ql-sdk\ql-config\quecproject\aboot\config\partition\CRANEL_SINGLE_QUEC_FLASH_LAYOUT_8M__GNSS_OPEN.json



强调一下，**调整的 size 大小必须是 4 的倍数**，不然 aboot 烧录固件时会报错。



Bin 文件的删除可以用 WinHex 软件去操作。

然后依次编译 .\build.bat app、.\build.bat firmware 即可。